Bohm then gave the example of atoms, the existence of which Mach resisted to the end, despite the overwhelming evidence to support them, because they couldn't be seen. Bohm made this point again shortly after he arrived in Brazil, in a letter he wrote to his friend and fellow physicist Arthur Wightman:

> Tentative concepts are needed, even before empirical evidence is available, to guide the choice and design of experiments, as well as to aid in their interpretation.… Very often, the actual empirical evidence for a new idea comes from surprising quarters (witness Brownean [*sic*] movement, the first evidence for existence of atoms, discovered by a biologist). However, such evidence can be appreciated only by people who are alert to the possibilities. For this reason, I would argue in favor of the widest possible diffusion of knowledge of *all* the possibilities among physicists. At a time like this, physicists ought to know of all the possibilities, and to feel that while they do not know which of these are correct, they must be ready, if necessary, to abandon even what seemed most secure and beautiful in the old point of view, in favor of what may seem arbitrary and ugly in the new point of view, if this should help explain something.

Yet, as Bohm pointed out in his 1952 papers, "[Positivism's] reflection still remains in the philosophical point of view implicitly adopted by a large number of modern theoretical physicists." To positivistically inclined physicists, it wasn't just that there was no need for a new interpretation of quantum physics—according to them, there was no need for any interpretation whatsoever. Quantum physics perfectly correlated and predicted observations, and that's all a scientific theory needed to do on a strictly positivist account of science. Any ideas about what nature was actually like that came along with a theory were just extra baggage. This was the logic behind Bohr's "rhetoric of inevitability," as the historian of science Mara Beller called it. Bohr and his followers said the Copenhagen interpretation was not just the correct way of understanding quantum physics—they said it was the only way to do it, the necessary and inevitable conclusion of the quantum revolution. "Every feature of [the Copenhagen interpretation]," claimed Léon Rosenfeld, one of Bohr's closest colleagues, "has been forced upon us as the only way to avoid the ambiguities which would essentially affect any attempt at an analysis in classical terms of typical quantum phenomena." Thus, according to Bohr's camp, the search